Revision – Exponential Functions, Finance and Trig

Grade 10 **August 2019**

WORK TO TWO DECIMAL PLACES WHERE NEEDED

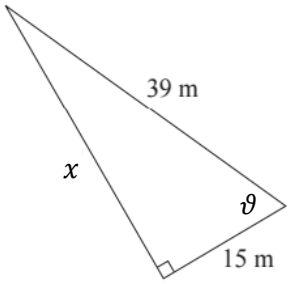
FORMULAE YOU MAY NEED:

$\boldsymbol{A = P\,(1 + i \times n)}$ $\boldsymbol{A = P(1 + i)^n}$

$\boldsymbol{A = \,(1 - i \times n)}$ $\boldsymbol{A = P(1 - i)^n}$

QUESTION 1

Use the following diagram to answer the questions which follow



a) Find the value of x, giving reasons (2)

b) Complete, leaving your answers in fractional form: **(3)**

$Sin\,\vartheta\ =$

$Cos\,\vartheta =$

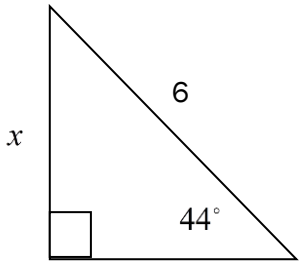
$Tan\,\vartheta =$

[5]

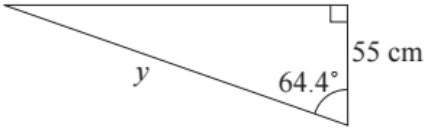
QUESTION 2

Solve for the unknown in the following:

a) **(3)**



b) **(3)**



[6]

QUESTION 3

a) If $\hat{A} = 155^\circ$ and $\hat{B} = 324^\circ$, use a calculator to determine the value of $\frac{\cos A}{2} + \sin 2B$ to two decimal places **(1)**

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b) $Tan^2\hat{A} =$ **(1)**

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c) $\sin\frac{B}{2} =$ **(1)**

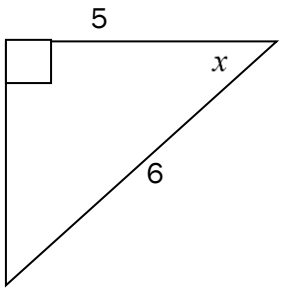
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[3]

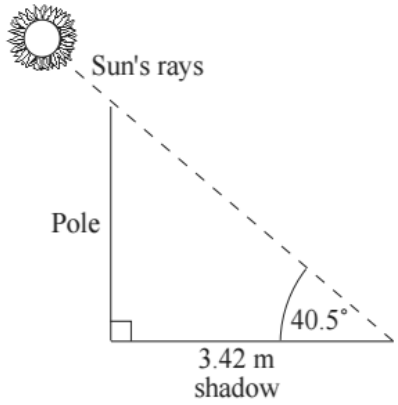
QUESTION 4

Solve for the unknown in the following:

a) **(3)**



b) The length of the shadow of a vertical pole is 3,42m long when the rays of the sun are inclined at an angle of $40{,}5^{\circ}$ to the horizontal. What is the height of the pole? Give the answer correct to two decimal places. **(3)**



[6]

QUESTION 8

Lungelo decides to deposit any money she earns over the next five years into a savings account. She opens an account and deposits R4 000 immediately. One year later she deposits another R2 000 into the account . Two years later she deposits a further R4300 into the account.
Calculate the total amount of money she has saved at the end of 5 years if the interest rate was 12% p.a. compounded quarterly.

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(6)